

The sealing system of Linear A: marks, circuits and iconography

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Summary

The Minoan archive writes in two systems: the tablets, which count, and the nodules, roundels and sealings, which identify — the distinction is Schoep's [4]. This paper describes the second with the procedure used for the first: measured distribution against a permutation null. Four results. **First**, the two systems share almost no vocabulary: of the 443 intact syllabic units, 389 occur only on tablets, 47 only on sealings and seven on both, against 34.4 expected under a null that preserves how often each unit occurs and how many units each document carries. **Second**, the reason: the sealing system does not write, it marks. Twenty-eight signs mark the 854 single-sign nodules of Haghia Triada, and five of them cover 83%. **Third**, mark and seal are paired: only 18 of the 42 identified seals serve more than one mark, against 24.4 under the null. **Fourth**, the marks fall into two circuits with their own dominant seal, and the two circuits are told apart by the iconography of those seals as well: one is 70% human figure and the other 70% animal. An edition artefact that came close to producing a false result is also declared, since it affects any analysis of sequence over this material.

Keywords: Linear A, nodules, seals, CMS, Neopalatial administration, null models.

1. The two systems share no vocabulary

Of the 443 intact syllabic units of the corpus, 389 occur only on tablets, 47 only on sealing supports, and **seven on both**: *409-VS, KA-KU-PA, PA-SE-JA, PA-TA-NE, *188-DU, KU-MI-NA-QE and SA-RA. A null that distributes the units across documents preserving how often each one occurs and how many each document carries (800 repetitions) expects **34.4 crossings**, with a minimum of 22. **p < 0.00125**.

This is not physical disconnection: the nodules were tied to the goods the tablets counted. It is that they name different things.

The seven that cross share a profile: they are parties whose goods leave a trace in both places. PA-TA-NE is the clean case, a list entry taking the figure 1 among other entries on HT 94b and HT 122a, and a name on nodule HT Wa 1019 of the same archive. PA-SE-JA carries 20 on the tablet and ½ on the sealing; KA-KU-PA opens HT 16 and stands beside cyperus on sealings.

2. The sealing system marks, it does not write

Of the 854 single-sign nodules of the GORILA edition [1] of Haghia Triada, **twenty-eight distinct signs mark them**, and five cover 83%: *301 (225), KU (151), KA (149), SI (92), RO (88). Against the 443 units the accounts use, the sealing system operates with twenty-eight marks. That accounts for the previous result: the intersection is small because the sealing repertoire is small.

3. Mark and seal are paired

Of the 42 seals identified on the nodules of Haghia Triada, **only 18 are used with more than one mark**, against 24.4 expected by reassigning the marks at random (400 repetitions, minimum of the null 21). $p < 0.0025$. Each seal attends to its mark.

4. Two circuits, told apart by what they depict

circuit	dominant seal	marks	impressions
I	AT 125	KU, KA, SI, RO, I, TA, DA-KA, SI-KA	455
II	AT 13	*301, ZE	312

Circuit I gathers eight marks, four of them among the five most frequent, with AT 125 dominant in every one (from 38% to 100%). Circuit II consists of *301, the most frequent mark of the archive with 271 sealed nodules, and ZE, both under AT 13.

The iconography separates them equally. The seal motifs are those of the CMS [2]. The dominant motif of circuit I is “man in kilt and man in robe walking left” (246 impressions); of circuit II, “bird flying right” (143). Aggregated by category, circuit I is 70% human figure and 3% animal; circuit II is 70% animal and 14% human. The difference statistic gives 684 against 3.5 under the null (maximum 21 over 500 reassignments), $p < 0.002$.

Two qualifications are filed with the result. The effect is carried in large part by one seal per circuit, so these are not two broad opposed repertoires but two principal seals of different class with their tail. And the motif classifier works on keywords and has gaps (stags did not fall in the animal list), so the direction of the result is firm but the exact percentages should not be cited as final.

5. An edition artefact worth declaring

Looking for sealing batches produced striking series: 225 consecutive nodules marked *301 (HT Wa 1623 to 1848), 151 with KU, 149 with KA. **They belong to the catalogue, not to the archive.** Of the 854 single-sign nodules, **825 consecutive pairs share a sign where 135 would be expected if they were mixed:** GORILA orders them by their mark, a hazard of the kind Packard [5] warned of. Any analysis of sequence, neighbourhood or batch over the nodules inherits that ordering and produces false results unless it declares it. It is set down here because the material is in common use and the artefact is not flagged in the edition.

6. Discussion

The archive of Haghia Triada sealed with two offices, told apart by their marks, by their seals and by the class of image those seals carry. The description does not depend on reading a single word, and it is the most complete Minoan administrative structure that has been measured.

What distinguishes the two offices in function remains open: whether they divide by commodity, by destination, by period or by rank. The nodule typology of Hallager [3] would decide what

distinguishes the two offices in function. The seal data carry their iconographic description but not their findspot within the site, and that is the information that would settle it.

One figure bears on the wider debate about reading Linear A: *301, the sign that the most recent Semitic proposals read as *na*, is the most frequent nodule mark of the archive, defines a circuit of its own under AT 13, and stands alone in 238 of its 288 attestations. Such a sign may also carry a syllabic value, but whoever assigns one must explain why that value leaves no trace in its dominant use.

References

[1] Godart, L. and Olivier, J.-P. 1976-1985. *Recueil des inscriptions en linéaire A* (GORILA), 5 vols. Paris: Geuthner. [2] Corpus der minoischen und mykenischen Siegel (CMS), Heidelberg. [3] Hallager, E. 1996. *The Minoan Roundel and Other Sealed Documents*. Liège: Aegaeum 14. [4] Schoep, I. 2002. *The Administration of Neopalatial Crete*. Salamanca. [5] Packard, D. W. 1974. *Minoan Linear A*. Berkeley: University of California Press.

Data and reproducibility

Corpus and code at github.com/BiomeMakers/kuro. Nodule and seal data derived from GORILA [1] IV-V [1] and from the CMS [2] through the LinearA Explorer database.

Statement of artificial intelligence assistance

The measurements, the null models and the drafting of this paper were carried out with the assistance of a language model (Claude, Anthropic), under the design, verification and responsibility of the author.